

ERES-CREDIT — Credit Division Under the Gate/Rating Split

Reference & Maintenance Specification

Instrument ID	ERES-CREDIT-2026-001
Spec version	Credit Spec v0.1 (DRAFT)
Reference implementation	<code>credit_split.py</code> v0.1
Companion instruments	BERA (<code>bera_rate.py</code> , ERES-BERA-2026-001) · EAAP (<code>eaap_proof.py</code>)
License	CCAL v2.1 (CARE Commons Attribution License)
Author / Lock authority	Joseph Allen Sprute, ERES Institute for New Age Cybernetics — ORCID 0000-0001-9946-3221
Method	H2C2H · MIEVM
Status	DRAFT for MIEVM audit. The weight vector is OPEN. Not deposited. Not load-bearing for any payout decision.

The asset, stated plainly. Dividing credit fairly is the mechanism that lets contribution become an asset others will tend rather than fight over. This instrument supplies the *structure* of that division — a vector, a held-state for the unverifiable, an authorization step, an auditable ledger. It deliberately does **not** supply the *valuation* (how much each kind of contribution is worth). That single number is a human lock, not a derivation, and it is the headline OPEN item below.

How to read this document.

- **To *apply* it:** read §1 (the split), §3 (the dimensions), §4 (the worked example).
- **To *understand* it:** add §2 (pipeline) and §5 (determinism).
- **To *maintain or extend* it:** §6 (lock status) and §7 (the maintenance contract) are what keep it fair over time. Do not change §7 without the lock authority.

1. The axiom: measuring credit and awarding it are different functions

This is the same wall BERA puts between rating and gate, applied to credit.

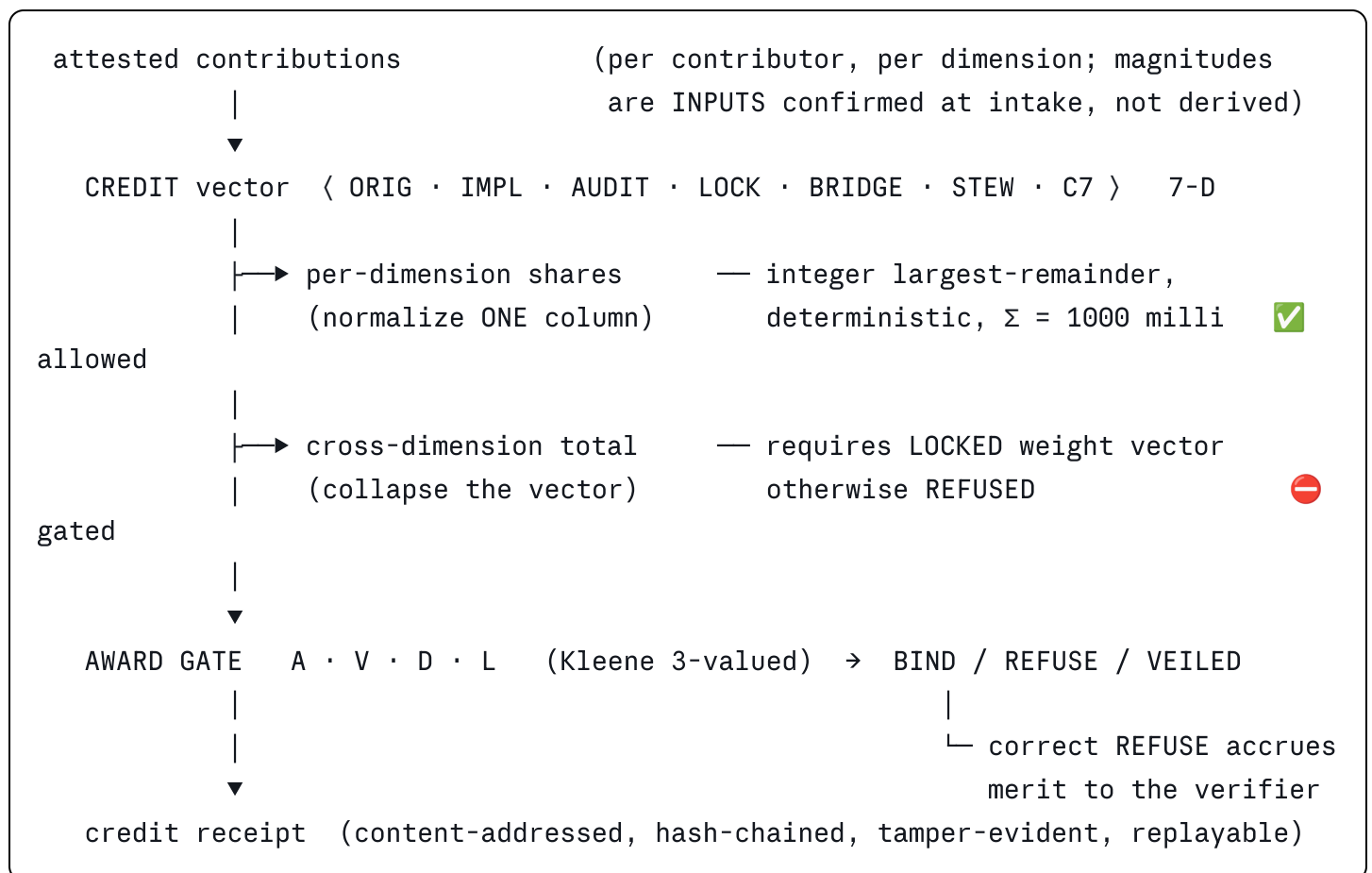
- **Measuring a contribution is a RATING** — the **CREDIT vector** (§3): continuous, multi-dimensional, "how much resonance did this add." Tuning-tier commentary.
- **Awarding credit is a GATE** — a bounded decision (**BIND**) / (**REFUSE**) / (**VEILED**) about whether an attribution, share, or payout is recognized. Auditable.

Two prohibitions follow, and they are exactly the two failure modes that wreck credit systems:

1. **A contribution measure must never silently become a reward.** There is always a named, bounded authorization step between "you did good work" and "you receive X."
2. **A credit decision must never be derived by thresholding a scalar-collapsed contribution.** This is the h-index trap, the citation-count trap, the single "impact score" trap. Credit is a **vector**; averaging it into one figure destroys the structure that makes the division defensible.

The reference implementation enforces both by construction: the CREDIT vector is never collapsed unless a **locked** weight vector exists, and no award is reachable except through the gate.

2. The pipeline at a glance



A contribution that cannot be verified is held **VEILED-C** — never given a fabricated value.

3. The CREDIT vector (the eggs)

Per contributor, a seven-dimensional vector. Six named dimensions plus a reserved seventh (C7) held VEILED-C, mirroring BERA's RATE. **Magnitudes per dimension are inputs** — attested at intake by the contribution record — not invented by this instrument.

Dim	Name	Credits	Status
ORIG	Origination	a novel construct or canonical form created	working
IMPL	Implementation	an executable realization	working
AUDIT	Audit	a fault found or corrected (the merit-for-refusal dimension)	working
LOCK	Canonical lock	an authority confirming a canonical form (a lock-authority act)	working
BRIDGE	Domain bridge	a cross-domain hypothesis or insight	working
STEW	Stewardship	maintenance, documentation, making it usable by others	working
C7	reserved	structural slot, held VEILED-C	dimension map OPEN

The named set is a *working* taxonomy. Adding, renaming, or splitting a dimension is a lock-authority action (it changes the dimension map, §6), and must preserve no-collapse and VEILED-C discipline.

3.1 Per-dimension shares (*computable, deterministic*)

Within a single dimension, shares are the contributors' magnitudes normalized to milli-units summing to exactly 1000, by **integer largest-remainder apportionment** with ties broken by contributor name. Normalizing one column is *not* a collapse of the vector, so this is permitted and fully deterministic.

$$\text{share}_i(d) = \text{floor}(\text{units}_i(d) \cdot 1000 / \sum \text{units}(d))$$
, then largest remainders receive +1 until the column sums to exactly 1000

A contributor **VEILED-C in that dimension is held out of the denominator** — their share is VEILED-C, and they neither receive a fabricated share nor dilute the resolved contributors.

3.2 Cross-dimension total (*refused without a LOCKED weight vector*)

Converting a contributor's whole vector to one number requires a weight vector **w_credit** across the dimensions. That weight vector is **OPEN** (the headline lock item). The implementation:

- **No locked weights** → the collapse is **REFUSED** (a `PermissionError`, by design).
- **Provisional weights supplied** → returns a result explicitly tagged `PROVISIONAL - PREVIEW (not for citation; weights OPEN)` — usable for internal tuning, never authoritative.
- **Locked weights** → returns an `AUTHORITATIVE` total.

This is the same discipline as ARI/ERI/RCI in BERA: the structure runs today; the canonical valuation waits for the lock authority.

3.3 VEILED-C — the held state for unresolvable credit

A contribution whose contributor or provenance cannot be verified is held **VEILED-C**: never coerced to a number, never given a share, never used to dilute others. This generalizes the attribution discipline already in practice (an unconfirmable party cited as a correspondent, not assigned a fictional stake) and echoes CyberRAVE — *Cybernetic Ratings Abolishing Veiled Exchanges*. When the evidence resolves, the dimension can be un-veiled — but only through the gate and the lock authority, never silently.

4. Worked end-to-end example

From the reference implementation's own run. Contributors are **roles** that map to named parties via attestation.

CREDIT vectors (inputs):

contributor	ORIG	IMPL	AUDIT	LOCK	BRIDGE	STEW	C7
originator	100	10	0	0	30	20	VEILED-C
implementer	5	80	40	0	0	25	VEILED-C
auditor	0	10	60	0	0	10	VEILED-C
lock-authority	0	0	0	100	0	5	VEILED-C
unverified-correspondent	0	0	0	0	VEILED-C	0	VEILED-C

Per-dimension shares (milli, Σ = 1000):

dimension	shares
ORIG	originator 952 · implementer 48
IMPL	originator 100 · implementer 800 · auditor 100
AUDIT	implementer 400 · auditor 600
LOCK	lock-authority 1000
BRIDGE	originator 1000 · <i>unverified-correspondent</i> <i>VEILED-C</i>
STEW	originator 333 · implementer 417 · auditor 167 · lock-authority 83

Note BRIDGE: the unverified correspondent is **held VEILED-C** and the originator still receives the full resolved share — the veiled party does not dilute the resolved one.

Cross-dimension collapse:

- authoritative total → **REFUSED** (weight vector OPEN)
- provisional preview with illustrative weights →

370

[PROVISIONAL-PREVIEW (not for citation; weights OPEN)]

Award gate (A · V · D · L):

case	A	V	D	L	decision
clean award	TRUE	TRUE	TRUE	TRUE	BIND
self-award	TRUE	TRUE	FALSE	TRUE	REFUSE
unverifiable provenance	TRUE	UNKNOWN	TRUE	TRUE	VEILED
no lock sign-off	TRUE	TRUE	TRUE	UNKNOWN	VEILED
forged attestation	FALSE	TRUE	TRUE	TRUE	REFUSE

Only the clean award reaches an actual award. The two correct REFUSEs accrue **merit 0.5 each** to the verifier — catching a bad claim is itself credited. The full ledger is hash-chained: flipping a recorded REFUSE to BIND is detected at the exact index, and replaying the same claims reproduces the chain head byte-for-byte.

5. Determinism

Credit records must be reproducible by any party from the same inputs.

- **Per-dimension apportionment** is integer largest-remainder with a deterministic tie-break (largest fractional part, then contributor name). No floating-point; shares sum to exactly 1000 on every platform.
- **Receipts** are content-addressed SHA-256 over a canonical (sorted-key) JSON encoding and hash-chained by `prev_receipt_id`. Replay of the same claim sequence reproduces every receipt id and the chain head.
- **The full 7-D vector is recorded in each receipt** — never a scalar.

6. Lock status — what is settled, what is open

Item	Status	Authority
Gate/rating split + two prohibitions (§1)	LOCKED	canonical axiom
CREDIT = 7-D vector; scalar collapse prohibited without locked weights	LOCKED	canonical
VEILED-C held-state discipline (§3.3)	LOCKED	canonical
Distinctness / no self-award (gate factor D; mirrors BERA Gate A)	LOCKED	canonical
Merit-for-refusal (correct REFUSE accrues to verifier)	LOCKED	canonical
Per-dimension apportionment = integer largest-remainder	LOCKED (v0.1 impl)	JAS
Hash-chained, replayable credit receipts	LOCKED (v0.1 impl)	JAS
Weight vector <code>w_credit</code> (cross-dimension valuation)	OPEN — the headline lock item	JAS
Named dimension set (ORIG/IMPL/AUDIT/LOCK/BRIDGE/STEW)	OPEN — working taxonomy	JAS
Formal C1...C7 dimension map	OPEN	JAS
Refusal-credit magnitude (currently 0.5)	OPEN	JAS

Item	Status	Authority
Intake rule for contribution magnitudes (who attests, how)	OPEN	JAS

The headline OPEN item is the weight vector. Until JAS locks `w_credit`, the instrument refuses to produce an authoritative cross-dimension total. This is intentional: the moment credit becomes money, people optimize for whatever dimensions you weight, so the weighting is an incentive design — a governance decision the lock authority must own and defend, not a number the math hands you.

7. The maintenance contract — how the division stays fair

Changing any of these is a **lock-authority action**, not a maintainer action. A reviewer should reject a pull request that violates any of them.

- **CI1 — Split.** A CREDIT measure never authorizes an award. An award never derives from a thresholded, scalar-collapsed credit.
- **CI2 — No collapse without a lock.** A cross-dimension total is REFUSED unless `w_credit` is LOCKED. Provisional previews are tagged non-authoritative and are not citable.
- **CI3 — VEILED-C discipline.** Unresolvable credit is held VEILED-C: never numbered, never given a share, never used to dilute resolved contributors. Un-veiling happens only through the gate and the lock authority.
- **CI4 — Distinctness.** No self-award: a contributor may not be the awarding authority for their own credit.
- **CI5 — Determinism.** Apportionment is integer largest-remainder with a fixed tie-break; replay of the same claims reproduces the chain head byte-for-byte.
- **CI6 — Merit-for-refusal.** A correct REFUSE accrues to the verifier; audit and review are first-class credited acts, not unpaid overhead.
- **CI7 — Ledger integrity.** Every award decision emits a hash-chained receipt recording the full 7-D vector. No receipt stores a single "credit score."
- **CI8 — Provisional ≠ canonical.** The weight vector, the dimension taxonomy, and any preview total are not citable until JAS lock.

Safe extension path. The instrument separates what is open from what is locked so the open parts can change without touching the load-bearing core:

- Adjusting the dimension taxonomy or the intake rule affects only the front-end; the split, VEILED-C, gate, and ledger are untouched.
- Locking the weight vector is a JAS action: confirm `w_credit`, move its row in §6 from OPEN to LOCKED, set `Creditweights.LOCKED`, and only then are authoritative cross-dimension

totals citable.

- Every change should be replay-verified and, ideally, fuzz-checked against CI1-CI7 before merge.
-

8. Position in the ERES instrument stack

1. **Deterministic reference implementation (EAAP gate)** — legally load-bearing (Daubert-relevant).
2. **MIEVM-LLM ensemble** — governance commentary only.
3. **BERA / ERES-CREDIT** — governance tuning only. CREDIT describes and apportions; it does not, by itself, move money or confer authority. The award gate records a recognized award; any consequence-bearing transfer flows through tier 1 discipline.

Any claim placing ERES-CREDIT in a load-bearing payout role without a locked weight vector and a named authorization step is out of specification.

Three Governing Principles (operationally binding): Don't hurt yourself. Don't hurt others. Build for generations to come.

ERES-CREDIT — Credit Division Under the Gate/Rating Split · ERES-CREDIT-2026-001 · CCAL v2.1